

MURAU, Austria

WMO: 112800

Lat: 47.111N

Lon: 14.177E

Elev: 2671

StdP: 13.33

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.6	(c) 11.2	(d) 3.0	(e) 7.1	(f) 10.1	(g) 6.4	(h) 8.4	(i) 14.6	(j) 11.3	(k) 31.5	(l) 9.8	(m) 31.6	(n) 2.3	(o) 290	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.8	(c) 84.5	(d) 66.1	(e) 81.3	(f) 64.7	(g) 78.0	(h) 63.5	(i) 67.7	(j) 81.3	(k) 66.0	(l) 78.4	(m) 64.4	(n) 75.7	(o) 6.4	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 63.1	(b) 95.5	(c) 72.9	(d) 61.5	(e) 90.0	(f) 70.9	(g) 60.0	(h) 85.5	(i) 69.2	(j) 33.6	(k) 81.7	(l) 32.2	(m) 78.6	(n) 30.9	(o) 76.0	(p) 77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.3	(b) 9.7	(c) 8.5	(e) 3.5	(f) 89.4	(g) 6.1	(h) 3.1	(i) -0.9	(j) 91.6	(k) -4.5	(l) 93.4	(m) -7.9	(n) 95.2	(o) -12.3	(p) 97.5		
			DB													
			WB													
				3.0												
				71.1												
				5.8												
				2.1												
				-1.2												
				72.6												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	26.9	31.7	38.9	46.4	54.5	61.1	64.2	63.6	55.2	47.6	37.7	28.8	
	DBStd	14.40	6.77	6.69	6.71	6.29	5.78	6.30	5.57	5.44	5.79	6.15	6.70	6.57	
	HDD50	2905	717	514	346	139	22	2	0	1	19	118	371	656	
	HDD65	6906	1182	934	810	558	327	147	82	92	295	539	819	1121	
	CDD50	1615	0	0	1	32	162	336	441	421	176	44	2	0	
	CDD65	140	0	0	0	0	2	31	58	47	2	0	0	0	
	CDH74	1705	0	0	0	5	78	431	641	513	37	0	0	0	
	CDH80	404	0	0	0	0	9	98	166	129	2	0	0	0	
Wind	WSAvg	3.3	2.9	3.2	3.7	3.8	3.9	3.7	3.6	3.4	3.2	3.0	2.7	2.6	
Precipitation	PrecAvg	38.7	1.3	1.4	1.9	2.3	3.6	5.1	5.7	5.4	3.7	3.5	3.0	1.8	
	PrecMax	46.5	2.6	4.4	4.5	3.6	5.0	6.8	11.5	9.1	5.3	8.8	7.2	4.4	
	PrecMin	32.6	0.4	0.2	0.4	0.6	1.7	3.0	1.5	3.0	1.9	0.2	0.1	0.4	
	PrecStd	4.2	0.7	1.0	0.9	0.7	0.9	1.1	2.1	1.7	0.9	1.8	1.7	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	56.5	65.3	73.8	81.0	86.7	89.0	88.1	78.5	70.3	59.1	45.9	
		MCWB	40.1	44.6	49.9	56.5	62.9	66.0	67.8	67.5	64.0	58.7	50.5	39.5	
	2%	DB	41.4	49.0	60.4	68.4	75.7	82.9	84.5	83.8	73.8	65.6	53.6	41.7	
		MCWB	37.0	40.4	47.4	53.5	60.1	64.4	66.8	66.4	61.3	55.4	47.5	37.7	
	5%	DB	38.0	44.8	55.9	64.3	71.9	79.3	80.8	79.8	70.3	61.8	50.0	39.1	
		MCWB	34.4	37.6	45.3	51.6	57.7	63.2	64.8	65.0	59.4	53.5	46.0	36.1	
	10%	DB	35.6	41.1	51.1	60.2	67.9	75.3	77.2	76.0	66.5	58.0	46.9	36.7	
		MCWB	32.8	35.7	42.6	49.3	55.8	61.9	63.7	63.3	57.7	51.4	43.9	34.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.9	45.2	50.9	57.3	65.0	69.0	70.2	69.8	64.8	59.4	51.7	41.4
			MCDB	45.1	54.9	63.0	71.5	76.1	80.3	84.5	84.9	76.0	68.2	56.5	43.9
2%		WB	37.6	41.0	48.1	54.3	61.0	66.2	67.9	67.3	62.6	56.6	48.6	38.5	
		MCDB	40.6	47.9	58.8	65.8	73.8	79.3	81.8	80.6	72.1	63.4	52.4	40.6	
5%		WB	35.3	38.3	45.9	52.1	58.8	64.3	66.0	65.6	60.5	54.4	46.6	36.6	
		MCDB	37.4	43.6	55.0	62.5	70.1	76.7	78.4	77.6	67.9	60.1	49.6	38.5	
10%		WB	33.3	36.2	43.3	50.0	56.6	62.4	64.2	63.9	58.5	52.3	44.2	34.6	
		MCDB	35.0	40.4	50.1	58.9	66.0	73.3	75.2	74.2	64.8	57.2	46.4	36.3	
Mean Daily Temperature Range		5% DB	MDBR	14.0	17.6	20.3	20.9	21.4	21.6	21.8	21.4	19.8	18.9	13.2	12.7
			MCDBR	15.6	23.2	29.6	29.9	29.4	29.5	28.5	28.3	26.0	24.6	17.6	14.4
	MCWBR		11.2	15.0	17.2	16.0	14.5	13.1	12.4	12.9	14.0	14.9	11.7	10.6	
	5% WB	MCDBR	14.0	21.2	27.5	27.1	27.5	26.8	26.2	26.0	22.9	22.6	15.7	12.8	
		MCWBR	10.4	14.2	16.5	14.9	14.1	12.6	12.2	12.5	12.8	14.0	10.9	9.8	
Clear-Sky Solar Irradiance	taub	0.278	0.309	0.355	0.401	0.412	0.418	0.418	0.411	0.377	0.348	0.315	0.280		
	taud	2.394	2.321	2.269	2.213	2.237	2.265	2.266	2.303	2.372	2.427	2.461	2.452		
	Ebn at Noon	259	271	273	270	271	269	267	264	263	254	242	246		
	Edh at Noon	22	29	36	42	43	42	41	38	32	26	21	19		
All-Sky Solar Radiation	RadAvg	403	679	1017	1319	1546	1710	1720	1525	1107	753	422	334		
	RadStd	48	89	121	151	172	137	144	171	142	79	66	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (98 neighbors)	+0.92	N/A	N/A	+1.49	+0.88	+0.70	-166	-247	+166	+99

Nomenclature: See separate page